FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VIII

JEE (Advanced)-2020

PAPER – 2

Time Allotted: 3 Hours **Maximum Marks: 186**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

1. ***Section-A (01 – 08, 19 – 26, 37 – 44)*** *contains 24 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*
   *Full Marks : **+4** If only (all) the four option(s) is (are) chosen.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen and both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*
   *Negative Marks : **–1** In all other cases.*

   Section-A (09 – 12, 27 – 30, 45 – 48) *contains 6 List-Match sets with 12 questions (each set has 2 questions). Each question has 4 statements in LIST-I & 6 statements in LIST-II. The codes for Lists have choices (A), (B), (C), (D) out of which only one is correct.*
   Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section-C (13 – 18, 31 – 36, 49 – 54)*** *contains 18 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

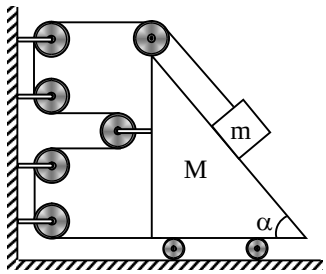
Physics

PART – I

SECTION – A
(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. If the acceleration of wedge in the figure shown is a m/s^2 towards left, then at this instant acceleration of the block (magnitude only) would be



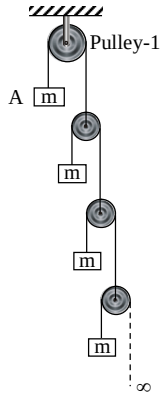
(A) $4a$ m/s^2

(B) $a\sqrt{17-8\cos\alpha}$ m/s^2

(C) $\sqrt{17a}$ m/s^2

(D) $\sqrt{17}\ \cos\frac{\alpha}{2} \times a$ m/s^2

2. System is released from rest. Only pulley 1 is fixed. Determine the acceleration of A



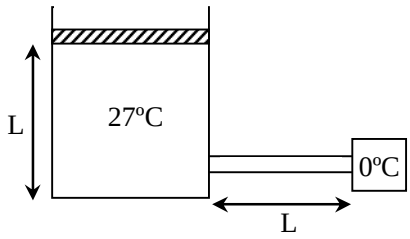
(A) $\frac{g}{2}$ downward

(B) $\frac{g}{2}$ upward

(C) $\frac{g}{4}$ downward

(D) $\frac{g}{4}$ upward

3. 0.5 mole of an ideal gas at constant temperature 27ºC kept inside a cylinder of length L and cross section area A closed by a mass less piston. The cylinder is attached with conducting rod length L cross section area $\left(\frac{1}{9}\right)$ m^2 and thermal conductivity K, whose other end is maintained at 0ºC. If piston is moved such that rate of heat flow through the conducting rod is constant then velocity of piston when it is at height $\frac{L}{2}$ from the bottom of cylinder [neglect any kind of heat loss from system)



(A) $\frac{K}{R}$ m/s

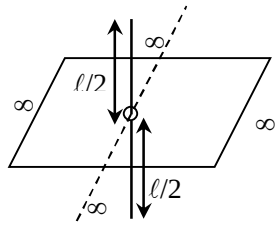
(B) $\frac{K}{10R}$ m/s

(C) $\frac{K}{100R}$ m/s

(D) $\frac{K}{1000R}$ m/s

Space for Rough work

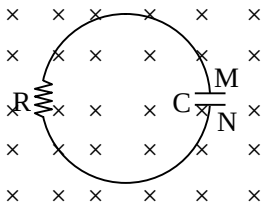
4. An infinite non conducting sheet having charge density σ and there is a small hole as shown in figure and there is rod of length λ having linear charge density λ and hinged in the hole as shown. If the mass of rod is m then time period of oscillation for small angular displacement



(A) $\pi\sqrt{\frac{m\in_0}{3\sigma\lambda}}$ (B) $2\pi\sqrt{\frac{2m\in_0}{3\sigma\lambda}}$

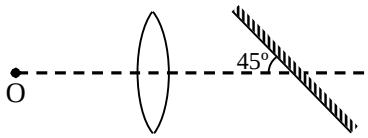
(C) $\frac{\pi}{2}\sqrt{\frac{m\in_0}{3\sigma\lambda}}$ (D) None of these

5. A uniform magnetic field decrease at a constant rate $\frac{dB}{dt} = -K$ where K is a positive constant. A circular loop of wire of radius 'a' containing a resistance R and a capacitance C is placed with its plane normal to the field



(A) maximum charge on the capacitor will be πa^2CK
(B) maximum charge on the capacitor will be $2\pi a^2CK$
(C) upper plate M, will be at higher potential when it is fully discharged
(D) lower plate N, will be at higher potential when it is fully discharged

6. A convex lens of focal length $\frac{a}{2}$ and a plane mirror are arranged as shown. The plane mirror is inclined at an angle 45º with the axis and the distance between lens and mirror is equal to focal length of the lens.



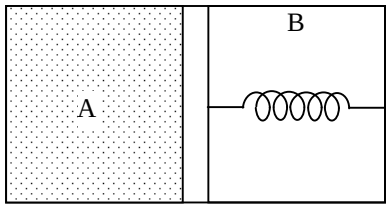
A point source is placed at a distance 'a' from lens at its axis. The distance between the source and the image formed after reflection from the mirror will be

(A) $\frac{a}{2}$ (B) $\frac{\sqrt{5}a}{2}$

(C) $\frac{\sqrt{10}a}{2}$ (D) None of these

Space for Rough work

7. A thermally insulated chamber of volume $2V_0$ is divided by a frictionless piston of area *S* into two equal parts *A* and B. Part A has an ideal gas at pressure P_0 and temperature T_0 and in part B is vacuum. A massless spring of force constant *k* is connected with piston and the wall of the container as shown. Initially spring is unstretched. Gas in chamber A is allowed to expand. Let in equilibrium spring is compressed by x_0. Then -



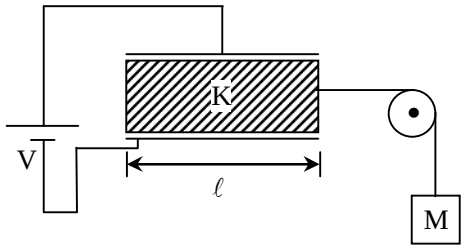
(A) final pressure of the gas is $\frac{kx_0}{S}$

(B) work done by the gas is $\frac{1}{2}kx_0^2$

(C) change in internal energy of the gas is $\frac{1}{2}kx_0^2$

(D) temperature of the gas is decreased

8. A capacitor with air as dielectric has capacity C as shown in figure is completely filled with dielectric slab of mass M and is connected to a block of mass M. The system is released form rest at time t = 0 when the capacitor is completely filled with dielectric. Then:
(Dielectric constant of slab = K)



(A) Charge on the plate of the capacitor when it is half filled with air is $\frac{(K+1)CV}{2}$

(B) The kinetic energy of the block when the capacitor is half filled with dielectric is $\frac{mg\ell}{4} - \frac{CV^2(K-1)}{8}$

(C) The potential difference between the plates when the capacitor is half filled with air is V

(D) The energy stored in the capacitor when it is half filled with dielectric is $\frac{K}{2}CV^2$.

Space for Rough work

(Matching List Type)

This section contains **TWO (02)** List-Match Sets. Each List-Match set has **TWO(02)** Multiple Choice Questions. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

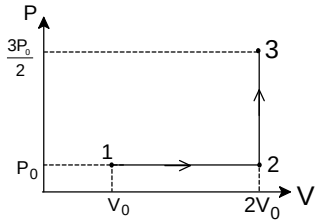
Answer the following by appropriately matching the list based on the information given in the paragraph.

In a thermodynamic process on an ideal monatomic gas, the infinitesimal heat absorbed by the gas is given by $T\Delta X$, where T is temperature of the system and ΔX is the infinitesimal change in a thermodynamic quantity X of the system. For a mole of monatomic ideal gas $X = \frac{3}{2}R\ \text{In}\left(\frac{T}{T_A}\right) + R\ \text{In}\left(\frac{v}{v_A}\right)$.

Here, R is gas constant, V is volume of gas, T_A and V_A are constants.
The List-I below gives some quantities involved in a process and List-II gives some possible values of these quantities.

	List – I		List – II
(I)	Work done by the system in process $1 \rightarrow 2 \rightarrow 3$	(P)	$\frac{1}{3}RT_0\ \text{In}2$
(II)	Change in internal energy in process $1 \rightarrow 2 \rightarrow 3$	(Q)	$\frac{1}{3}RT_0$
(III)	Heat absorbed by the system in process $1 \rightarrow 2 \rightarrow 3$	(R)	RT_0
(IV)	Heat absorbed by the system in process $1 \rightarrow 2$	(S)	$\frac{4}{3}RT_0$
		(T)	$\frac{1}{3}RT_0(3+\text{In}2)$
		(U)	$\frac{5}{6}RT_0$

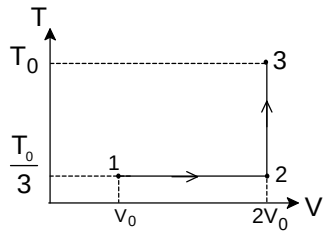
9. If the process carried out on one mole of monatomic ideal gas is as shown in figure in the PV diagram with $P_0V_0 = \frac{1}{3}RT_0$. The correct match is



(A) I – Q,II – S,III – R,IV – U
(B) I – S,II – R,III – Q,IV – T
(C) I – Q,II – R,III – P,IV – U
(D) I – Q,II – R,III – S,IV – U

Space for Rough work

10. If the process carried out on one mole of monatomic ideal gas is as shown in the TV – diagram with $P_0V_0 = \frac{1}{3}RT_0$. The correct match is



(A) I – P,II – R,III – T,IV – P
(B) I – P,II – R,III – T,IV – S
(C) I – S,II – T,III – Q,IV – U
(D) I – P,II – T,III – Q,IV – T

Answer the following by appropriately matching the list based on the information given in the paragraph.

A musical instrument is made using four different metal strings, 1, 2, 3 and 4 with mass per unit length μ, 2μ, 3μ and 4μ respectively. The instrument is played by vibrating the strings by varying the free length in between the range L_0 and $2L_0$. It is found that in string-1 (μ) at free length L_0 and tension T_0 the fundamental mode frequency is f_0.
List-I gives the above four strings while list-II lists the magnitude of some quantity.

	List – I		List - II
(I)	String – 1 (μ)	(P)	1
(II)	String – 2 (2μ)	(Q)	½
(III)	String – 3 (3μ)	(R)	$1/\sqrt{2}$
(IV)	String - 4 (4μ)	(S)	$1/\sqrt{3}$
		(T)	3/16
		(U)	1/16

11. If the tension in each string is T_0, the correct match for the highest fundamental frequency in f_0 units will be
(A) I – Q, II – S, III – R, IV – P
(B) I – P, II – R, III – S, IV – Q
(C) I – P, II – Q, III – T, IV – S
(D) I – Q, II – P, III – R, IV – T

Space for Rough work

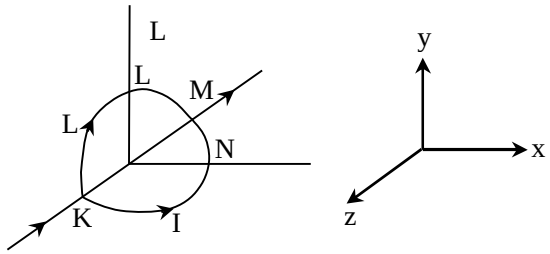
12. The length of the strings 1,2,3 and 4 are kept fixed at L_0, $\frac{3L_0}{2}, \frac{5L_0}{4}$ and $\frac{7L_0}{4}$, respectively. Strings 1,2,3 and 4 are vibrated at their 1st, 3rd, 5th and 14th harmonics, respectively such that all the strings have same frequency. The correct match for the tension in the four strings in the units of T_0 will be

(A) I – P, II – R, III – T, IV – U
(B) I – P, II – Q, III – T, IV – U
(C) I – T, II – Q, III – R, IV – U
(D) I – P, II – Q, III – R, IV – T

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded - off to the **second decimal place**; e.g. XXXXX.XX).

13. A circular loop of radius *R* is bent along a diameter and given a shape as shown in the figure. One of the semicircles (KNM) lies in the *x-z* plane and the other one (KLM) in the *y-z* plane with their centres at the origin. Current I is flowing through each of the semi circles as shown in figure.



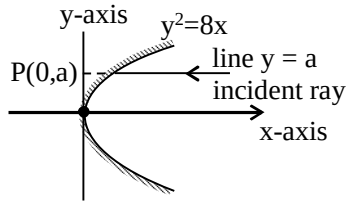
A particle of charge *q* is released at the origin with a velocity $\vec{v} = v_0\hat{i}$.

Find the magnitude of instantaneous force $\vec{F}$ on the particle if $\mu_0 q v_0 I = 8R$. Assume that space is gravity free.

14. The electric potential between a proton and an electron is given by, $V = V_0 \ln\left(\frac{r}{r_0}\right)$ where V_0 and r_0 are constants and *r* is the radius of the electron orbit around the proton. Assuming Bohr's model to be applicable, it is found that *r* is proportional to n^x, where *n* is the principal quantum number. Find the value of *x*.

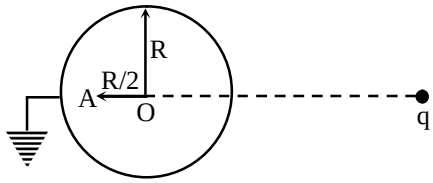
Space for Rough work

15. Figure shows a parabolic reflector in x-y plane given by $y^2 = 8x$. A ray of light traveling along the line y = a is incident on the reflector. Find where the ray intersects the x-axis after reflection.

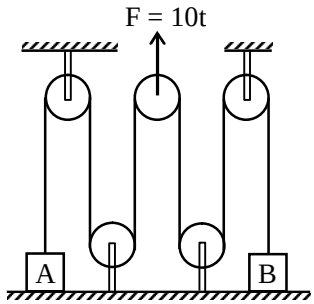


16. A hydrogen like atom (atomic number Z) is in a higher excited state of quantum number n. This excited atom can make a transition to the first excited state by successively emitting two photons of energies 10.20 eV and 17.00 eV respectively. Alternatively, the atom from the same excited state can make a transition to the second excited state by successively emitting two photons of energies 4.25 eV and 5.95 eV respectively. Determine the value of Z. (Ionisation energy of hydrogen atom is 13.6 eV).

17. A point charge q is placed at a distance 3R from the centre of a earthed conducting shell of radius R as shown in figure
Then find magnitude electric potential at point A, R/2 distance away centre, due to charge induce on the shell.
If $q = 28\pi\varepsilon_0 R$.



18. In the arrangement shown $m_A = 1$ kg and $m_B = 2$ kg while all the pulleys & strings are massless & frictionless. At $t = 0$, a force $F = 10$ t starts acting over central pulley in vertical direction. When *B* loses contact with floor the velocity of *A* is (5n) m/sec. Find out the value of '*n*'.



Space for Rough work

Chemistry

PART – II

SECTION – A
(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. For the reaction: $A + B \longrightarrow Product$; following data is given

Expt. No.	[A]	[B]	Rate(M/sec)
1	0.02 M	0.03 M	4×10^{-3}
2	0.04 M	0.06 M	1.6×10^{-2}
3	0.01 M	0.06 M	4×10^{-3}

Which of the following is correct?
(A) Given reaction may be elementary
(B) The value of rate constant is 6.67 $M^{-1}sec^{-1}$
(C) Given reaction may be complex
(D) If volume of container is doubled in first experiment, rate becomes 10^{-3} M/sec

20. Which of the following statement(s) is(are) correct regarding $H_4P_2O_6$?
(A) It contains P-P linkage
(B) It contains P-O-P linkage
(C) Its basicity may be 3
(D) The number of $p\pi - d\pi$ bonds = 2

21. Select the correct statement(s).
(A) The total number of Bravais lattice in 2-D crystal system is 5
(B) A cubic system contains 9 plane of symmetry
(C) For monoclinic system, $a \neq b \neq c$ and $\alpha \neq \beta \neq \gamma \neq 90^0$
(D) The packing fraction of BCC is 52%

22. A quantity of 720 gm water is added in 230 gm ethanol at a certain temperature to get 1 L of solution. Which of the following is/are correct regarding the solution formed?
(A) The density of solution is 950 kg/m^3
(B) The mole fraction of ethanol is 0.11
(C) The molarity of solution is 5 M
(D) The molality of solution is 6.94 m

Space for Rough work

23. The correct statement(s) for the following addition reaction is(are)

(i) $\text{Trans}-2-\text{butene} \xrightarrow{Br_2/CHCl_3} M \text{ and } N$

(ii) $\text{Cis}-2-\text{butene} \xrightarrow{Br_2/CHCl_3} O \text{ and } P$

(A) (**M** and **O**) and (**N** and **P**) are two pairs of enantiomers
(B) Bromination proceeds through trans-addition in both the reactions
(C) **O** and **P** are identical molecules
(D) (**M** and **O**) and (**N** and **P**) are two pairs of diastereomers

24. Which of the following are not redox reactions

(A) $Mg + N_2 \longrightarrow Mg_3N_2$

(B) $K_4[Fe(CN)_6] + H_2SO_4 + H_2O \longrightarrow K_2SO_4 + CO + FeSO_4 + (NH_4)_2SO_4$

(C) $I_2 + 3Cl_2 \longrightarrow ICl_3$

(D) $CuSO_4 + NH_3 \longrightarrow [Cu(NH_3)_4]SO_4$

25. Consider a parallel reaction,

$$(P) \xrightarrow[E_a = E]{k_2} \begin{cases} \xrightarrow[E_a = E_1]{k_1} (X) \\ \xrightarrow[E_a = E_2]{k_2} (Y) \end{cases}$$

Which of the following statement(s) is/are true?

(A) When both the reactions are of 1st order, $\frac{k_1}{k_2} = \frac{[X]}{[Y]}$

(B) $\frac{k_1}{k_2} = \frac{[X]}{[Y]}$ is true for all parallel path reactions irrespective of their order.

(C) The overall energy of activation is given by $E = \frac{k_1E_1 + k_2E_2}{E_1 + E_2}$, if the two parallel reactions are of same order.

(D) For first order reaction, the plot of $\ln t_{1/2}$ vs $1/T$ is a straight line and slope of the line is equal to E/R (where R is gas constant).

26. By passing 1F of electricity

(A) 1 mol of Zn deposited
(B) 12 g of Mg deposited
(C) 1/3 mol of Al deposited
(D) 11.2 L of O_2 at NTP evolved

Space for Rough work

(Matching List Type)

This section contains **TWO (02)** List-Match Sets. Each List-Match set has **TWO(02)** Multiple Choice Questions. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

Answer the following by appropriately matching the list based on the information given in the paragraph.

Match the solutions obtained by mixing different volumes of equimolar acid HA and base BOH given in list–I with the rise in temperature given in list–II if the solution obtained by mixing 10 ml each of the same acid and base shows an elevation in temperature of 5°C.

LIST - I		LIST - II (Rise in temperature)	
(I)	100 ml of HA + 100 ml of BOH	**P.**	5°C
(II)	10 ml of HA + 20 ml of BOH	**Q.**	4°C
(III)	20 ml of HA + 30 ml of BOH	**R.**	3.3°C
(IV)	50 ml of HA + 150 ml of BOH	**S.**	0.66°C
		T.	2.5°C
		U.	2.6°C

27. Which of following is correct combination.
(A) I – P & II – R (B) II – P & III – R
(C) III – S & I – P (D) IV – P & I – Q

28. Which of following is incorrect combination.
(A) I – P & III – Q (B) I – P & II – R
(C) III – Q & IV – T (D) III – S & IV – Q

Space for Rough work

Answer the following by appropriately matching the list based on the information given in the paragraph.

LIST - I		LIST - II	
(I)	$\underset{5\text{ ml }1N}{CH_3COOH} + \underset{5\text{ ml }1N}{NaOH}$	**P.**	pH > 7
(II)	$\underset{15\text{ ml }1N}{CH_3COOH} + \underset{10\text{ ml }1N}{NaOH}$	**Q.**	pH < 7
(III)	$\underset{5\text{ ml }1N}{HCl} + \underset{15\text{ ml }1N}{NH_4OH}$	**R.**	Buffer
(IV)	$\underset{1\text{ ml }1N}{HCl} + \underset{1\text{ ml }2N}{NaOH}$	**S.**	Hydrolysis occurs
		T.	pH < 8
		U.	pH > 8

29. Which of following is correct combination
(A) I – P & II – Q, R (B) II – P, R & III – Q
(C) III – S & I – P (D) IV – P & I – Q

30. Which of following is incorrect combination.
(A) I – P & III – R (B) I – P & II – R
(C) II – Q & III – Q (D) I – P & IV – P

Space for Rough work

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded - off to the **second decimal place**; e.g. XXXXX.XX).

31. 8 gm of methane is burnt with 4.48L of O_2 at STP. Find out the volume of CO_2 gas produced at STP and also the weight of CO_2 gas.

32. The pK_a of the carboxyl group in an amino acid valine, $(CH_3)_2CHCH(NH_2)(COOH)$ is 2.31 and the pK_a for the amino group of the same amino acid is 9.69. Compute the isolectric point (pH_i) for valine and draw the structure of this amino acid when the pH of the solution equals to pH_i. Also draw the structures of valine that predominate at pH=2 and pOH= 2.

33. The relative rate of abstraction per hydrogen atom of 3°H, 2°H and 1°H of alkane during photo chlorination is 5.0 : 3.8 : 1. The percentage of the major product formed in the chlorination of 2,2–dimethyl butane is x% then the value of x is

34. What is the percent by mass of iodine needed to reduce the freezing point of benzene to 3.5°C? The freezing point and cryoscopic constant of pure benzene are 5.5°C and 5.12 K/m respectively.

35. A diatomic molecule has a dipole moment of 1.2D. If its bond distance is 1.0Å. What fraction of an electronic charge exist on each atom?

36. Calculate the standard e.m.f. of the reaction $Fe^{3+} + 3e^- \rightarrow Fe_{(s)}$ ε_3^0 = ?
Given $Fe^{3+} + e^- \rightarrow Fe^{2+}$ ε_1^0 = 0.771V $Fe^{2+} + 2e^- \rightarrow Fe_{(s)}$ ε_2^0 = – 0.44V

Space for Rough work

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. $f(x) = \cos\pi(|x| + [x])$, then

(A) f is continuous at x = 1/2
(B) f is continuous at x = 0
(C) f is differentiable in (-1, 0)
(D) f is differentiable in (0, 1)

38. Tangent are drawn to the circle $x^2 + y^2 = 50$ from a point 'P' lying on the x – axis. These tangents meet the y – axis at points 'P_1' and 'P_2'. Possible coordinates of 'P' so that area of triangle PP_1P_2 is minimum, is/are

(A) (10, 0)
(B) $\left(10\sqrt{2}, 0\right)$
(C) (-10, 0)
(D) $\left(-10\sqrt{2}, 0\right)$

39. Two particle start from the same point (2, -1), one moving 2 units along the line x + y = 1 and the other 5 units along x – 2y = 4. If the particles move towards increasing y, then their new position will be

(A) $\left(2-\sqrt{2}, \sqrt{2}-1\right)$
(B) $\left(2\sqrt{5}+2, \sqrt{5}-1\right)$
(C) $\left(2+\sqrt{2}, \sqrt{2}+1\right)$
(D) $\left(2\sqrt{5}-2, \sqrt{5}-1\right)$

40. If $\frac{x}{a} + \frac{y}{b} = 1$ is a tangent to the curve $x = 4t$, $y = \frac{4}{t}$, $t \in R$ then

(A) $a > 0, b > 0$
(B) $a > 0, b < 0$
(C) $a < 0, b > 0$
(D) $a < 0, b < 0$

Space for Rough work

41. All values of $x \in \left(0, \frac{\pi}{2}\right)$ such that $\frac{\sqrt{3}-1}{\sin x}+\frac{\sqrt{3}+1}{\cos x}=4\sqrt{2}$ are

(A) $\frac{\pi}{15}$ (B) $\frac{\pi}{12}$

(C) $\frac{11\pi}{36}$ (D) $\frac{3\pi}{10}$

42. Let a, b, c be distinct complex numbers with $|a|=|b|=|c|=1$ and z_1, z_2 be the roots of the equation $az^2+bz+c=0$ with $|z_1|=1$. Also let P and Q represents the complex numbers z_1 and z_2 in the complex plane with $\angle POQ=\theta$ where O is the origin then

(A) $b^2=ac, \theta=\frac{2\pi}{3}$ (B) $\theta=\frac{2\pi}{3}, PQ=\sqrt{3}$

(C) $b^2=ac, PQ=2\sqrt{3}$ (D) $2b^2=ac, \theta=\frac{\pi}{3}$

43. If the sum of the first 100 terms of an AP is -1 and the sum of the even terms lying in the first 100 terms is 1, then

(A) common difference is $\frac{3}{50}$ (B) first term is $\frac{149}{50}$

(C) common difference is $\frac{7}{50}$ (D) first term is $-\frac{151}{50}$

44. A line segment is divided so that the lesser part is to the greater part is equal to the greater part is to the whole. If R is the ratio of the lesser part of the greater part, then

(A) $R^2+R-1=0$ (B) $R^2+R^{-1}=2$

(C) $R^{\left(R^2+R^{-1}\right)}+R^{-1}=2$ (D) $R^2+R+1=0$

Space for Rough work

(Matching List Type)

This section contains **TWO (02)** List-Match Sets. Each List-Match set has **TWO(02)** Multiple Choice Questions. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

Answer the following by appropriately matching the list based on the information given in the paragraph.

List – I		**List - II**	
(I)	Two sides of traingle are given by the roots of the equation $x^2 - 5x + 3 = 0$. If the angle between these sides is $\pi/3$, then the product of inradius and circumradius of the triangle is	**(P)**	12
(II)	In a triangle ABC, if the radii of ex – circles r_1, r_2 and r_3 are given by $r_1 = 8$, $r_2 = 12$ and $r_3 = 24$, then value of the side a is	**(Q)**	15
(III)	Let a be a positive integer not exceeding 10, then the probability that the equation $x^2 - 2(a^2 - 1)x + 2a^2 - 7a + 3 = 0$ has one positive and one negative roots is	**(R)**	$\frac{3}{7}$
(IV)	Let (α, β) be a point from which two perpendicular tangents can be drawn to the ellipse $4x^2 + 5y^2 = 20$. If $F = 4\alpha + 3\beta$, then the maximum value of F	**(S)**	$\frac{1}{5}$
		(T)	$\frac{2}{3}$
		(U)	$\frac{1}{3}$

45. Which of following is correct combination
(A) I – T & II – P
(B) II – P & III – R
(C) III – S & I – P
(D) IV – P & I – Q

Space for Rough work

46. Which of following is incorrect combination.
(A) I – Q & III – Q
(B) I – P & II – R
(C) III – Q & IV – P
(D) III – S & IV – Q

Answer the following by appropriately matching the list based on the information given in the paragraph.

List – I		List - II	
(I)	The coefficient of two consecutive terms in the expansion of $(1+x)^n$ will be equal, then n can be	**(P)**	9
(II)	If $15^n + 23^n$ is divided by 19, then n can be	**(Q)**	10
(III)	${}^{10}C_0\ {}^{20}C_{10} - {}^{10}C_1\ {}^{16}C_{10} - \ldots\ldots$is divisible by 2^n, then n can be	**(R)**	11
(IV)	If the coefficient of T_r, T_{r+1}, T_{r+2} terms of $(1+x)^{14}$ are in A.P., then r is less than	**(S)**	12
		(T)	14
		(U)	16

47. Which of following is correct combination
(A) I – P & II – Q, R
(B) II – P, R & III – Q
(C) III – S & I – P
(D) IV – P & I – Q

48. Which of following is incorrect combination.
(A) I – Q & III – Q
(B) I – P & II – R
(C) I – P & IV – T
(D) II – Q & III – Q

Space for Rough work

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded - off to the **second decimal place**; e.g. XXXXX.XX).

49. Given (2 + 6 + 10 + 14 + ……+ 2a) + (2 + 6 + 10 + 14 + …….+ 2b) = (2 + 6 + 10 + 14 + ……+ 2c), where a + b + c = 21 and a > 6. Then a – b is equal to _____.

50. Seven digit numbers are formed using digits 1, 2, 3, 4, 5, 6, 7, 8, 9 without repetition. The probability of selecting a number such that product of any 5 consecutive digits is divisible by either 5 or 7 is p. Then 12p is equal to _____.

51. If a singular matrix $A = \left[a_{ij}\right]_{2\times 2}$ always commute with $B = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$ such that $\frac{a_{11}}{a_{12}} = \sqrt{k}$, then k is equal to_____.

52. If the function $f(x) = 2x^3 - 9ax^2 + 12a^2x + 1$ has a local maximum at $x = x_1$ and a local minimum at $x = x_2$ such that $x_2 = x_1^2$, then a is equal to _____.

53. If positive numbers x, y, z are in A.P, then the minimum value of $\frac{x+y}{2y-x} + \frac{y+z}{2y-z}$ is equal to____.

54. The number of odd proper divisors of 24300 is N + 11, then N is equal to _____.

Space for Rough work